Priya Raghunathan

Senior Software Developer — AI and Machine Learning Engineering

Toronto, Canada | priya.raghunathan@example.com | linkedin.example.com/in/priya-raghunathan | github.example.com/priyar-ml

PROFESSIONAL SUMMARY

Senior machine learning engineer with 12 years of experience across data science, applied machine learning, and production LLM systems in insurance, retail, and telecom analytics. Builds end-to-end ML systems in Python and PyTorch, from data pipelines and training to evaluation, serving, and monitoring. Recent work focuses on large language models: retrieval-augmented generation (RAG), document understanding, LoRA fine-tuning of open-weight models such as Gemma and Llama, and AI agents with tool calling. Automated a majority of insurance claim intake with an extraction pipeline that reached 0.93 field-level F1, and reduced LLM inference cost by 38% through distillation and efficient serving. Works closely with product managers, legal, and claims experts on model governance, and mentors engineers moving from data science into ML engineering.

TECHNICAL SKILLS

Programming languages: Advanced: Python, SQL. Working knowledge: Scala, TypeScript, Bash.

AI and machine learning: Advanced: large language models (LLMs), RAG, prompt engineering, fine-tuning (LoRA, QLoRA), NLP, document understanding, classification, forecasting, recommendation systems. Working knowledge: computer vision.

Frameworks and libraries: PyTorch, Hugging Face Transformers, PEFT, vLLM, LangChain, LlamaIndex, scikit-learn, LightGBM, XGBoost, TensorFlow, spaCy, pandas.

Models and model APIs: Open-weight models (Gemma, Llama, Mistral), hosted LLM APIs including Vertex AI Gemini and OpenAI, embedding models, cross-encoder rerankers.

MLOps and infrastructure: MLflow, Airflow, Kubeflow Pipelines, Docker, Kubernetes, GPU autoscaling, Evidently drift monitoring, FastAPI model services.

Data and retrieval: PostgreSQL with pgvector, Elasticsearch (BM25), BigQuery, feature stores.

Cloud platforms: Google Cloud (Vertex AI, GKE, BigQuery), AWS (SageMaker, S3).

Responsible AI and governance: Evaluation harnesses, golden datasets, red-teaming, PII redaction, model cards, human-in-the-loop review.

PROFESSIONAL EXPERIENCE

Northlake Insurance Analytics (insurtech) — Toronto, Canada

Senior Machine Learning Engineer — *Apr 2022 – Present*

- Led the design of a claims document understanding pipeline that combines OCR, a layout-aware transformer, and LLM extraction validated against a JSON schema; it automates 62% of first-notice-of-loss intake with field-level F1 of 0.93 on a held-out set of 4,800 documents.
- Built a retrieval-augmented generation assistant for claims adjusters over 1.2 million policy documents, using hybrid retrieval (BM25 and dense embeddings in pgvector) with a cross-encoder reranker; the grounded-answer rate on a 600-question evaluation set rose from 71% to 89%.
- Created the team’s LLM evaluation harness with golden datasets, regression gates in CI, red-team prompt suites, and PII redaction checks; no prompt or model change ships without it.
- Reduced inference cost by 38% by distilling a large hosted model’s outputs into a LoRA fine-tuned 7B open-weight model served with vLLM on autoscaled GPU nodes, keeping extraction F1 within 0.01 of the larger model.
- Mentor three ML engineers; co-author model cards and risk assessments with legal and claims.

Tessellate Retail AI (retail technology) — Toronto, Canada

Machine Learning Engineer — *Jul 2018 – Mar 2022*

- Rebuilt store-level demand forecasting with LightGBM and calendar, promotion, and weather features, reducing weighted MAPE from 18.0% to 13.5% across 900 grocery stores.
- Built training and feature pipelines with Airflow, MLflow, Docker, and Kubernetes, cutting the time from experiment to production model from six weeks to eight days.
- Developed a two-tower recommendation model in TensorFlow for personalized promotions; an online A/B test showed a 6.2% lift in click-through rate.

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- Added drift and data-quality monitoring with Evidently, which caught two upstream schema changes before they reached production forecasts.

Ravel Analytics (analytics consulting) Bangalore, India
Data Scientist *Aug 2014 – Jun 2018*

- Built NLP models with scikit-learn and spaCy to route telecom customer complaints into 24 categories, reaching 87% routing accuracy and reducing manual triage effort by about 40%.
- Developed churn prediction models with logistic regression and XGBoost for two telecom clients and presented retention recommendations to their marketing teams.
- Standardized the team's Python project template with unit tests and reproducible environments, used across 15 client engagements.

SELECTED PROJECTS

EvalForge — LLM evaluation toolkit (open source)
Technologies: Python, pytest, Pydantic, DuckDB, Streamlit

- **Problem:** Teams compared prompts and models in ad hoc spreadsheets, so quality regressions reached releases.
- **Contribution:** Designed the test-case format, scoring plugins (exact match, JSON schema validity, calibrated LLM-as-judge), and CI integration.
- **Result:** Used by four teams at Northlake; caught 23 regressions before release in its first year.

Bilingual claims extraction with open-weight models
Technologies: Gemma and Llama models, Hugging Face PEFT (LoRA), vLLM, FastAPI

- **Problem:** French-language claims had lower extraction accuracy than English claims.
- **Contribution:** Assembled a 6,000-document bilingual training set with claims experts, fine-tuned and compared open-weight models, and built the serving endpoint.
- **Result:** French field-level F1 improved from 0.84 to 0.91 with no drop in English accuracy.

Underwriting triage agent (prototype)
Technologies: Python, LLM function calling, LangChain, PostgreSQL

- **Problem:** Underwriters spent hours collecting data from five systems before making a decision.
- **Contribution:** Built a tool-calling AI agent that gathers data and drafts a cited summary for human approval.
- **Result:** In a four-week pilot with 12 underwriters, median preparation time fell from 55 to 20 minutes.

EDUCATION

Lakeshore University of Toronto Toronto, Canada
Graduate Certificate in Artificial Intelligence (part-time) *2019 – 2020*

- Coursework: deep learning, natural language processing, probabilistic graphical models.

Deccan Institute of Technology Bangalore, India
B.Tech. Computer Science and Engineering *2010 – 2014*

- First-class honours; final-year project on sentiment analysis of product reviews.
- Coursework: machine learning, statistics, algorithms, database systems.

HACKATHONS, CONFERENCES, AND PROFESSIONAL ACTIVITIES

- **Toronto Applied ML Day (2025)** — Speaker: "Evaluating LLM extraction in regulated industries", on golden datasets, CI gating, and human review.
- **Northern Lights AI Hackathon, Toronto (2024)** — Team lead; built a 24-hour AI agent prototype that summarized municipal permit applications using an open-weight Gemma model with retrieval.
- **DocAI Workshop (2023)** — Co-author of a workshop paper on measuring field-level accuracy in document extraction systems.

CERTIFICATIONS AND ADDITIONAL QUALIFICATIONS

Certifications (illustrative): Cloud ML Engineering, Lakeshore Professional Institute (2021).

Mentoring: Mentor in the Northlake ML guild; reviewer for student ML projects.

Languages: English (fluent), Tamil (native), Hindi (professional).

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